Lynbrook Math Summer Camp 2026
C5 Expected Value

Lynbrook Math Club

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C5.1 Lecture

What is a Random Variable?

A **random variable** is just a number whose value depends on the outcome of some random process. Formally, it's a function that assigns a number to each possible outcome.

Random variables come in two flavors:

- **Discrete**: takes on finitely (or countably) many values. E.g. $X =$ the number shown when you roll a die, or $Y =$ the number of heads in three coin flips.
- **Continuous**: takes on a whole continuum of values, e.g. $Z =$ a uniformly random point along a 12-unit board, or the location a dart lands on a board.

For a discrete random variable, expected value is the familiar weighted sum $\sum p_i x_i$. For a continuous one, we replace "probability of a value" with "area (or length) of a region relative to the whole".

What is Expected Value?

If a random variable X takes value x_i with probability p_i, its **expected value** (also called its *mean*, written $\mathbb{E}[X]$) is

$$\mathbb{E}[X] = \sum_i p_i x_i.$$

You should think of $\mathbb{E}[X]$ as the long-run average value of X if you repeated the random experiment over and over. Two things to keep in mind right away:

- $\mathbb{E}[X]$ does *not* have to be a value that X can actually take! The expected value of a single die roll is 3.5, even though a die never lands on 3.5.
- The probabilities p_i must sum to 1.

Linearity of Expectation

The single most useful tool in this entire handout is **linearity of expectation**:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y],$$

and more generally,

$$\mathbb{E}[X_1 + X_2 + \cdots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n].$$

this is always true, whether or not $X_1, \ldots, X_n$ are independent, and no matter how ugly or dependent on each other they are. Whenever a complicated random variable can be written as a sum of simpler pieces, you can find its expected value piece by piece and add. Contrast this with a rule like $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, which is only true when X and Y *are* independent.

A couple of other easy but handy facts, for a constant c:

$$\mathbb{E}[cX] = c\,\mathbb{E}[X], \qquad \mathbb{E}[X + c] = \mathbb{E}[X] + c.$$

Indicator Random Variables

The best way to exploit linearity is with **indicator random variables**: a variable I that equals 1 if some event A happens, and 0 otherwise. Then

$$\mathbb{E}[I] = 1 \cdot P(A) + 0 \cdot P(\text{not } A) = P(A).$$

So whenever a problem asks for the *expected number of things* with some property (matches, adjacent pairs, fixed points, successes, ...), try writing that count as a sum of indicator variables, one per “thing” that could have the property, and just add up their probabilities. You never need to worry about how the indicators depend on one another because of linearity of expectation.

Problem-Solving Strategy

When a problem mentions (or secretly wants) an expected value, try, roughly in this order:

1. Compute $\sum p_i x_i$ directly. This only works when there aren’t too many cases.
2. Write the quantity as a sum of simpler random variables and apply linearity.
3. If you’re counting occurrences of some property, use indicator variables.